The Electromagnetic Form Factor of the ρ-meson on the Lattice

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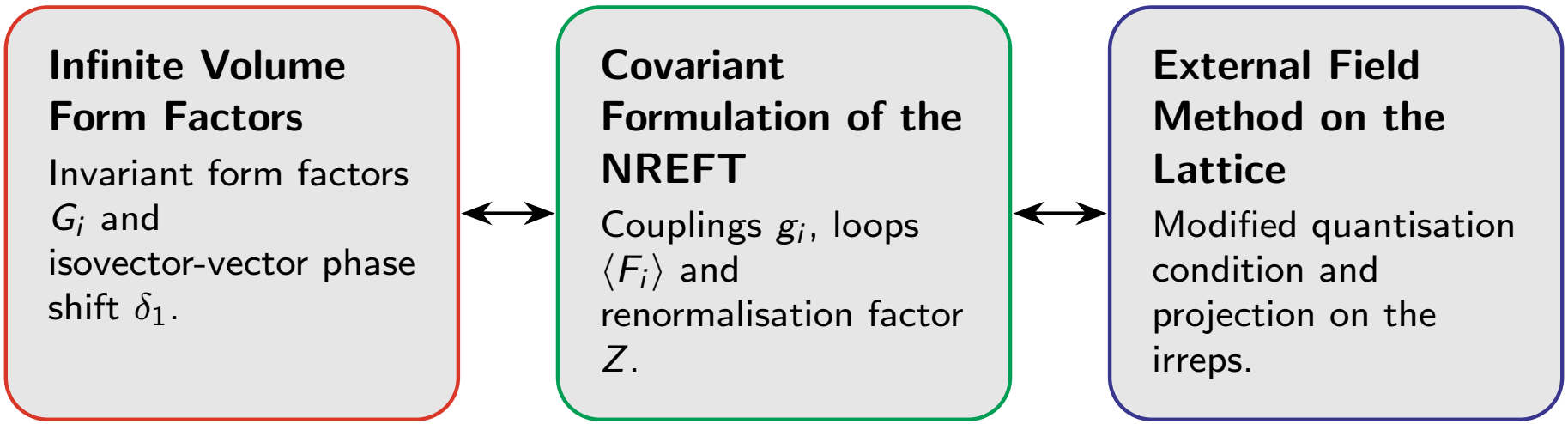






In a Nutshell

- The theory of vector resonances (specifically the ρ-meson) in the covariant formulation of the NREFT.
- A framework to consistently extract infinite volume form factors of unstable particle from lattice data.
- At the same time, a detailed analysis of the NREFT quantities and ChPT with an explicit inclusion of the ρ-meson.

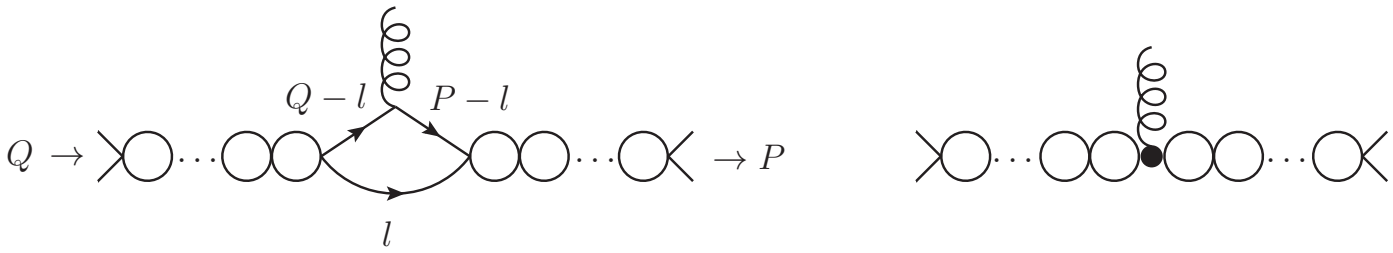


Form Factor Fundamentals

- In the infinite volume, given a quantum field with the quantum numbers of the ρ-meson obeying relevant constraints:
  - $\rightsquigarrow$ The two-point function of the composite ρ-meson field develops a simple pole on the second Riemann sheet at $s = s_R$.
  - $\rightsquigarrow$ The three-point function involving the electromagnetic current develops a double pole at $s = s' = s_R$.
  - $\rightsquigarrow$ The extraction of the EM form factor corresponds to calculating the residue of the three-point function.
- The matrix element of the EM current is conventionally decomposed in terms of three linearly independent invariant form factors:

$$\begin{aligned}\langle P, s|J^{\alpha}(0)|Q, s'\rangle =& \tilde{\varepsilon}_{\mu}(P, s)\,\big(-G_1(k^2)g^{\mu\nu}(P+Q)^{\alpha} \\ &- G_2(k^2)(k^{\mu}g^{\nu\alpha} - k^{\nu}g^{\mu\alpha}) \\ &+ \frac{1}{4s_R}G_3(k^2)k^{\mu}k^{\nu}(P+Q)^{\alpha}\big)\,\varepsilon_{\nu}(Q, s').\end{aligned}$$

- A direct extraction of the form factors of unstable states on the lattice poses significant challenges.
- The contributions can be split into a triangle diagram and the contact term:



(a) (b)

- The triangle diagram does not exhibit a regular dependence on the box size L. (*Bernard et al., 2012*)

- To overcome this, we make use of the external field method. (*Lozano et al., 2022*)
  - ⇝ The system is placed in a periodic external field, which is coupled to a given current.
  - ⇝ The modified Lüscher approach lets one extract relevant physical results.
- In this work, we discuss the conceptual challenges associated with the calculation of the ρ-meson electromagnetic form factor.
- We do not concentrate on the technical details of a lattice implementation.

The Form Factor in the NREFT Framework

- We work with a covariant NREFT Lagrangian that describes the elastic $\pi^+\pi^0$ scattering with a dipion field Ψ that models the ρ-meson: (*Müller et al., 2022, 2023*)

$$\mathcal{L} = \overbrace{\sum_{a=+,0} \phi_a^\dagger 2W(i(v\cdot\partial) - W)\phi_a}^{\text{Pion Lagrangian}} - \overbrace{\Psi_\mu^\dagger \Psi^\mu}^{\text{Rho Lagrangian}}$$

$$W = \sqrt{M^2 + \partial^2 - (v\cdot\partial)^2}$$

quantisation axis

$$\Psi^\mu = \phi_+ \overset{\leftrightarrow}{W}{}^\mu f(\overset{\leftrightarrow}{W}{}^2)\phi_0,$$

$$W^\mu = v^\mu W + i(\partial^\mu - v^\mu (v\cdot\partial)),$$

$$f(\overset{\leftrightarrow}{W}{}^2) = f_0 + f_1 \overset{\leftrightarrow}{W}{}^2 + \dots$$

- The function f is related to the $\pi\pi$ scattering phase shift in the p-wave:

$$q_0 \cot \delta_1(q_0) = -\frac{24\pi\sqrt{P^2}}{q_0^2 f^2(q_0^2)}$$

- The wavefunction renormalisation factor Z is also related to experimental input through

$$Z = \frac{8q_R^2}{1 + 2b\sqrt{-q_R^2}},$$

$$b = \frac{d}{dq_0^2}[q_0 \cot \delta_1(q_0)]\Big|_{q_0^2 = q_R^2},$$

where $q_0^2 = \frac{s}{4} - M_\pi^2$.

Inclusion of Electromagnetic Interaction

- To study the electromagnetic form factor, we need to gauge the NREFT Lagrangian.
- This is not straightforward due to the presence of non-local operators.
- Under minimal coupling, we have

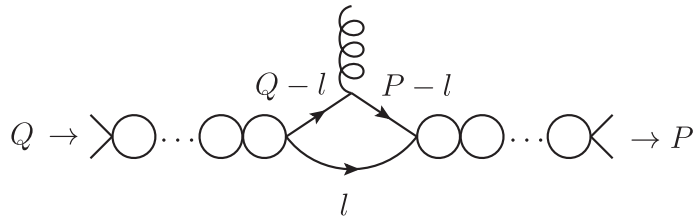
$$W^2 = M^2 + \partial^2 - (v \cdot \partial)^2 \to W^2 + B + \mathcal{O}(e^2).$$

Then it can be shown that

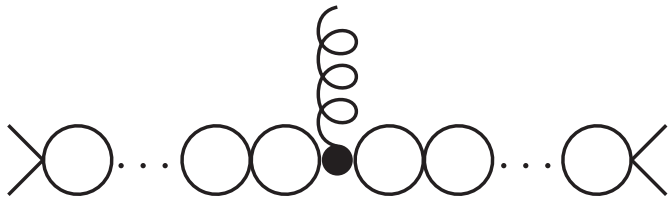
$$W \to W + D, \quad D = \int_0^\infty dt\ e^{-tW} B e^{-tW},$$

which can be evaluated in the momentum space.

- This directly gives us the two pion photon vertex:



- This leads to a contribution to the matrix element of the EM current between the two pion states in terms of loop integrals.
- Further, gauging the derivative terms in the rho Lagrangian gives us the four-pion photon vertex:



- However, to make the matching apparent, we include the relevant terms manually:

$$\mathcal{L}_{\text{c.t.}} = \sum_{i=1}^{3} g_i e \mathscr{O}_i,$$

where

$$\mathscr{O}_1 = \Psi_\mu^\dagger \{\partial^\beta F_{\alpha\beta}, W_\alpha\} g^{\mu\nu} \Psi_\nu,$$
$$\mathscr{O}_2 = \Psi_\mu^\dagger i F^{\mu\nu} \Psi_\nu,$$
$$\mathscr{O}_3 = \Psi_\mu^\dagger \frac{1}{2} \{(\partial^\mu F^{\nu\alpha} + \partial^\nu F^{\mu\alpha}), W_\alpha\} \Psi_\nu,$$
$$\mathscr{O}_4 = \Psi_\mu^\dagger [W^\mu, [W^\nu, \{\partial^\beta F_{\alpha\beta}, W_\alpha\}]] \Psi_\nu.$$

- This then gives a contribution to the matrix element of the EM current between the two pion states in terms of the couplings g_1, g_2 and g_3.
- For $P^2 = Q^2$, the operators $\mathscr{O}_3$ and $\mathscr{O}_4$ lead to the same amplitude up to an overall factor: we set $g_4 = 0$.

The Invariant Form Factors

- Finally, we can relate the infinite volume invariant form factors to the NREFT quantities:

$$
\begin{aligned}
G_1(k^2) &= -eZ(g_1 k^2 + \langle F_1 \rangle), \\
G_2(k^2) &= -eZ(g_2 + \langle F_2 \rangle), \\
\frac{1}{4s_R} G_3(k^2) &= eZ(g_3 + \langle F_3 \rangle),
\end{aligned}
$$

  where the $\langle F_i \rangle$'s are loop integrals.

- It can be verified that $Z\langle F_1 \rangle$ is normalised to unity at zero momentum transfer since G_1 is normalised due to Ward identity.

- At this point, we can estimate the order of magnitudes of the couplings g_i.
- Matching to $\mathcal{O}(p^4)$ ChPT Lagrangian and making use of Schenk parametrisation:

$$g_2 = -\frac{M_\pi^2(2l_1^r - l_2^r)}{24\pi F_\pi^4 a_1^1} + \text{ loops} = 2.8 \cdot 10^{-2} M_\pi^{-2},$$
$$g_1 = g_3 = 0 + \text{ loops}.$$

- The loop contribution is expected to be of the same order.
- A detailed numerical analysis of the matching process and an alternate estimate of the couplings based on an explicit inclusion of a ρ-meson field in ChPT is also in progress. (*Shamanauri et al.*)

The Form Factor in a Finite Volume

- We consider the following setup: the ρ-meson system is placed in a time-independent periodic electromagnetic field with a period ω on a lattice of size L.
- One can then extract the invariant form factors at $k^2 = -\omega^2$.
- Suppose the ρ-meson is stable: the form-factor and the external field induces corrections to the interaction Hamiltonian, diagonalising which gives us the invariant form factors (*Standard Lüscher approach*).

Modified Lüscher Approach in an External Field

- In the case of an unstable ρ-meson: we know that the two-point function of a charged pion in a finite volume is modified in the presence of an external field.
- The finite volume spectrum is determined by the poles of the short-range contributions.
- The quantisation condition is given by

$$\det(\Pi^{-1}_{\mu\nu} - D^{0}_{\mu\nu}) = 0$$

"self-energy" the two-point function of the interpolating field

- Now one can consider specific background field configurations and project on to the relevant irreps. This enables one to fit the couplings g_i.

Summary

- The covariant formulation of the NREFT for the ρ-meson has been derived:
  - $\rightsquigarrow$ This lets one derive the matrix element of the EM current in terms of the couplings g_i and loop integrals.
  - $\rightsquigarrow$ On the one hand, this can be related to infinite volume invariant form factors.
  - $\rightsquigarrow$ On the other hand, the couplings g_i can be fit to lattice data.
- This lets one extract the EM form factor of the ρ-meson from the lattice.
- Additionally, we also provide a crude estimate of the couplings g_i from ChPT.
- A detailed analysis of the matching to ChPT is also being studied.

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Thanks for your attention!